MIL-I-8505
6 JANUARY 1953
Superseding
AN-I-51
30 November 1948

MILITARY SPECIFICATION

INDICATORS; TEMPERATURE, ELECTRICAL-RESISTANCE,
-50° to +300°C

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers electrical-resistance type thermometer indicators of the size shown on Drawing AN5792, covering the temperature range of -50° to +300°C.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standard, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-M-151	Metals; General Specification for Inspection of

Military

MIL-C-5015	Connectors; Electrical
MIL-C-6055	Containers; Metal, Reusable, Interior Shipping
MIL-D-5028	Drawings and Data Lists; Preparation of (For Engines, Accessories and other Auxiliary Equipment)
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-N-3336	Nut, Self Locking, Instrument Mounting
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-P-6906	Plates, Information and Identification
MIL-S-7742	Screw Threads, Standard, Aeronautical

Air Force-Navy Aeronautical

AN-L-1	Luminescent Material; Fluorescent

STANDARDS

MIL-STD-129	Marking of Shipments

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DRAWINGS

Air Force-Navy Aeronautical Standard Drawings

AN5792	Indicator - Temperature, Electrical-Resistance, -50° to +300°C
AND10398	Metals - Definition of Dissimilar
AND10400	Numerals and Letters - Aircraft Instrument Dial, Standard Form of

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall conform to applicable specifications and shall be as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended. In the construction of the equipment, noninflammable material shall be employed to the greatest extent. The use of lightweight materials and weight-saving designs is a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.1.1 Metals.- Metals shall be of the corrosion-resistant type, unless suitably protected to resist corrosion during normal service life.

3.1.1.1 Dissimilar metals.- Dissimilar metals are defined by Drawing AND10398. Unless suitably protected against electrolytic corrosion, by means of protective coatings or hermetic sealing, intimate contact of dissimilar metals shall be avoided.

3.1.2 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.1.3 Fungus-Proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. External materials which are nutrients for fungi shall be treated with a fungicidal agent, as approved by the procuring activity.

3.1.4 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.4.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.2 Design and construction.- The indicator shall conform to Drawing AN5792. The design shall be reasonably simple to permit overhaul or repair without requiring numerous special tools and fixtures. The indicator shall be so constructed as to withstand the normal strains of jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.2.1 Indicator case.- The case shall conform to Drawing AN5792. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.2.1.1 The case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.2.1.2 The case shall be filled with helium of at least 98-percent purity, free of dust particles, and containing not more than 0.006 milligrams of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the helium in the case shall be approximately 1 atmosphere.

3.2.2 Cover glass.- The cover glass shall be suitable for hermetic sealing and shall be clear, flat, and free from flaws which interfere with normal reading of the indicator.

3.2.3 Pointers.- The pointer shall be as light as practicable and sufficiently rigid to prevent flexing under vibration. The pointer design shall conform to figure 1. The pointer shall be channeled to provide adequate strength.

3.2.3.1 Pointer overswing.- Pointer stops, if used, shall be located to permit the pointer to overswing not less than 3 percent of full scale deflection beyond each end of the scale.

3.2.3.2 Pointer overlap.- The pointer length shall be such that the pointer tip will extend into the scale a distance equal to 1/4 to 1/2 of the length of the shortest graduation. The pointers shall not touch each other at mid-scale.

3.2.4 Dial.- The dial shall conform to figure 1, and shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated.

3.2.4.1 Scale.- The scale of the indicator shall be as long as practicable and shall not be less than 1-7/8 inches in length as measured on the outer arc of the minor graduations.

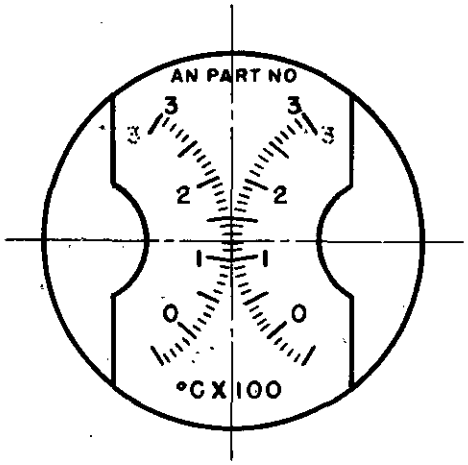
3.2.5 Dial and cover glass location.- The distance between the inside surface of the cover glass and the dial shall not exceed 0.140 inch.

3.2.6 Mask.- A thin metal mask finished in durable dull black conforming to figure 1 shall be provided behind the cover glass to cover all moving parts of the mechanism except the pointer and to provide for a uniform visible length of pointer regardless of pointer indication. The minimum length of pointer visible shall be 1/2 inch.

3.2.7 Weight.- The weight of the complete indicator shall not exceed 2.0 pounds.

3.2.8 Balancing.- The moving element of the indicator shall be balanced by an approved means (such as threaded lock or split nuts, or a wire helix moving along a cross arm and counterweight arm) with easily accessible means for rebalancing. The use of solder, shellac, or similar means for holding balance weights will not be acceptable because it prevents ready rebalancing.

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ALTERNATE LOCATION
FOR NUMERALS SHOWN
IN PHANTOM LINE

DUAL



POINTER

MARKING	HEIGHT OR LENGTH ±.010	WIDTH OF LINE ±.005	MATERIAL OR FINISH
Numerals 0, 1, 2, and 3	0.156	0.020	
50° Graduations	0.156	0.020	Fluorescent-Luminescent Material
Lettering "°C x 100"	0.125	0.020	
Visible portion of pointer	- - -	- - -	
10° Graduations	0.094	0.010	
AN Part No.	0.063	0.010	Durable Dull Black
Background of dial and mask	- - -	- - -	

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:
DECIMALS ±.010

FIGURE 1. Pointer and dial

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3.2.9 Bearings and pivots.- Moving elements shall turn on hardened, accurately ground, and highly polished pivots bearing on highly polished first-quality sapphire bearings or other materials that have been approved as alternates to precious gems.

3.2.10 Soldering.- Only noncorrosive fluxes shall be used for soldering unless corrosion agents and products have been satisfactorily removed.

3.2.11 Wiring.- The internal indicator wiring shall be of tinned flexible stranded wire, wherever practicable.

3.2.12 Electrical receptacle.- Each indicator shall be equipped with an electrical receptacle in accordance with Specification MIL-C-5015 as shown on Drawing AN5792. The receptacle shall be wired as follows:

Terminal	Connection
A	+24-Volt battery
B	Resistance bulb (left)
C	Common - negative 24-volt battery and remaining resistance bulb connections
D	Resistance bulb (right)

3.2.13 Calibration.- The indicator shall be calibrated for the resistance bulb temperature versus resistance characteristics given in table I.

3.2.14 Voltage.- The indicator shall be designed to operate normally from 28.5 volts, dc, ±0.5 volts.

3.3 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.4 Finish.- Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used

3.4.1 Metals.- Metals shall be suitably protected to meet the requirements of this specification.

3.4.2 Indicator case.- The bezel ring, if used, and the mounting flange shall be durable dull black.

3.5 Screw threads.- Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.6 Performance.- Tolerances, limits, and accuracy requirements under various operating conditions shall be as specified in Section 4.

3.7 Markings.- All markings shall be durable to withstand usage encountered in service.

3.7.1 Dial and pointer.- The dial and pointer shall be marked and finished as shown on figure 1. The form of the numerals and letters shall conform to Drawing AND10400. Luminescent material shall conform to Specification AN-L-1.

3.8 Identification of product.-

3.8.1 Identification.- The AN part number shall be marked on the dial in durable dull black. The location need not be as shown on figure 1, but it shall be in a location which is readily visible from the front of the indicator.

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TABLE I

Test tolerances

Test Point	Test Resist-ance	Scale error at room temperature	Reduced voltage	Scale error at low temperature	Scale error at high temperature	Scale error at room temperature after open and short circuit
		Tolerances ±	Tolerances ±	Tolerances ±	Tolerances ±	Tolerances ±
°C	Ohms	°C	°C	°C	°C	°C
-50	37.95	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
0	50.00	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
+50	63.55	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
100	79.00	5.0 (5.0)	4.0 (4.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
150	96.50	4.0 (4.0)	4.0 (4.0)	7.0 (7.0)	7.0 (7.0)	7.0 (7.0)
200	116.50	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
250	139.00	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)
300	164.50	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)	7.0 (7.0)

Test point	Test resist-ance	Scale error at room temperature after Vibra-tion failure test	Extreme temperature exposure			Scale error at room temperature after salt spray
			Scale error at low temp	Scale error at room tem-perature after low temp	Scale error at room tem-perature after high temp	
		Tolerances ±	Tolerances ±	Tolerances ±	Tolerances ±	Tolerances ±
°C	Ohms	°C	°C	°C	°C	°C
-50	37.95	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)
0	50.00	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)
+50	63.55	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0((5.0)	12.0 (5.0)
100	79.00	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)
150	96.50	12.0 (10.0)	15.0 (7.0)	12.0 (4.0)	12.0 (4.0)	12.0 (4.0)
200	116.50	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)
250	139.00	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)
300	164.50	12.0 (10.0)	15.0 (10.0)	12.0 (5.0)	12.0 (5.0)	12.0 (5.0)

Note: Tolerances not inclosed in parentheses are applicable only when the complete sequence of Qualification tests is conducted. Tolerances inclosed in parentheses are applicable only when the specific Qualification test is conducted independent of sequence.

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3.8.2 Nameplate.- A nameplate or nameplates, conforming to Specification MIL-P-6906, shall be securely attached to the indicator case and shall be legibly and permanently marked with the information listed below. The information marked in the spaces provided on the nameplate shall be in accordance with Specification MIL-M-7911.

INDICATOR - TEMPERATURE, ELECTRICAL-RESISTANCE
AN Part No.
24 volts, dc
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's name or trade-mark
US Property

3.8.3 Use of AN or MIL designations.- AN or MIL designations shall not be applied to a product, except for Qualification test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification, or from the Aeronautical Standards Group.

3.9 Installation.-

3.9.1 Mounting screws.- The contractor shall furnish sufficient mounting screws for installing the indicator. They shall be No. 6-32, NC-2, round head, brass machine screws, having durable dull black oxidized or black nickel finish. The length shall be sufficient for mounting on a panel 3/16 inch thick.

3.9.2 Mounting nuts.- When required by the AN part number as shown on Drawing AN5792, a sufficient number of removable No. 6-32, NC-2 spring nuts of a type conforming to Specification MIL-N-3336 to fit the mounting lugs of the instrument, shall be furnished.

3.9.3 Envelope.- An envelope furnished by the contractor containing mounting screws, installation instructions, and removable spring nuts, shall be packaged with each instrument and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTALLATION
INSTRUCTIONS, MOUNTING SCREWS, AND SPRING NUTS"

3.10 Workmanship.- All parts shall be manufactured and processed in accordance with high-grade aircraft instrument manufacturing practice.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of tests.- The inspection and testing of indicators shall be classified as follows:

(a) Qualification tests: Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product.

(b) Inspection tests: Inspection tests are those tests accomplished on indicators manufactured and submitted for acceptance under contract.

4.2 Qualification tests.-

4.2.1 Sampling instructions.- The Qualification test samples shall consist of two indicators of each manufacturer's model number upon which qualification is desired.

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Samples shall be forwarded to the Supply Officer, Naval Air Material Center, US Naval Base Station, Philadelphia 12, Pennsylvania, Attention: Superintendent of the Instrument Laboratory, plainly identified by securely attached durable tags, marked with the following information:

Sample for Qualification test
INDICATOR; TEMPERATURE, ELECTRICAL-RESISTANCE, -50° to +300°C
AN Part No.
Manufacturer's Part No.
Name of manufacturer
Submitted by (name) (date) for Qualification test in accordance with the requirements of Specification MIL-I-8505 under authorization (reference authorizing letter dated _____)

4.2.2 Tests.- The Qualification tests of the indicators shall consist of all the tests of this specification as described under "Test methods." The Qualification tests may, at the option of the qualifying activity, be supplemented with tests under actual service conditions.

4.3 Inspection tests.- The Inspection tests shall consist of Individual tests, Sampling tests, and Special tests.

4.3.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.3.2 Individual tests.- Each indicator submitted for acceptance under contract shall be subjected to the following tests described under "Test methods":

(a) Examination of product
(b) Scale error at room temperature
(c) Position error
(d) Friction error
(e) Insulation resistance
(f) Sealing

In addition, the indicators shall be subject to any of the other tests specified herein that the Inspector considers necessary to determine conformance with the requirements of this specification.

4.3.3 Sampling tests.- The Inspector shall select at random, for tests, at least five indicators from each 500 indicators, or less, of each lot which have passed the required Individual tests. A lot shall consist of indicators of the same part number submitted for inspection under the same contract or order. Each sample selected shall be subjected to the following tests described under "Test methods":

(a) Reduced voltage operation
(b) Scale error at low temperature
(c) Scale error at high temperature
(d) Open and short circuit
(e) Resistance bulb current
(f) Vibration error
(g) Vibration failure
(h) Magnetic effect

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4.3.4 Special tests.- Three of the inspection samples representing the manufacturer's first contract or order, and samples selected as often thereafter as the Inspector deems advisable, shall be subjected to any or all of the following tests described under "Test methods":

(a) Response time
(b) Damping factor
(c) Power consumption
(d) Extreme low temperature exposure
(e) High temperature exposure
(f) Altitude
(g) Thermal shock
(h) Salt spray
(i) Mounting lugs

4.3.5 Rejection and retest.- When any representative sample fails to meet the requirements of the Sampling tests, an additional representative sample test lot shall be selected and the test repeated. If failure occurs in the second lot selected, the complete lot shall be rejected and returned at the contractor's expense. Any indicator failing to meet the requirements of the Individual tests shall be rejected and returned at the contractor's expense. Indicators which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and action taken to correct the original defects shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.4 Test conditions.-

4.4.1 Atmospheric conditions.- Unless otherwise specified, all tests required by the specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and a temperature of approximately 25°C (77°F). When tests are made with atmospheric pressure or temperature substantially different from the above values, proper allowance for change in indicator reading shall be made for the difference from the specified conditions.

4.4.2 Tapping.- Unless otherwise specified, the indicator shall be tested in normal operating position (plane of dial vertical) and shall be lightly tapped before a test reading is taken.

4.4.3 Test voltage.- Unless otherwise specified, the supply voltage applied shall be 28.5 ±0.5 volts direct current.

4.4.4 Vibration.- A vibration stand shall be used which will vibrate at any desired frequency between 500 and 3,000 cpm and shall subject the indicator to vibration such that a point on the case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.5 Test methods.-

4.5.1 Examination of product.- Each indicator shall be examined externally to determine conformance with the specification with respect to material, workmanship, marking, and envelope.

4.5.2 Scale error at room temperature.- The resistance values specified in table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied, and the scale errors shall be determined at the test points in table I. The scale errors at any of the test points shall not exceed the tolerances specified in table I.

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4.5.3 Position error.-

4.5.3.1 No voltage.- With no voltage applied, the change in pointer indication produced by rotating the indicator from the normal operating position, 90 degrees clockwise and counterclockwise about the longitudinal axis (dial in vertical plane) or with the plane of the dial horizontal shall be noted. The pointer shall remain off scale at the low temperature end of the scale in the above described positions.

4.5.3.2 Voltage.- The test specified for "No voltage" shall be repeated with voltage and test resistance applied. This test may be performed at any indication at the discretion of the Inspector. The maximum change in indication, at any of the above described positions, from the normal position shall not exceed 7.0°C. When the voltage to the indicator is switched off, the pointer shall return to the off-scale position without tapping.

4.5.4 Friction error.- Differences in indication shall be noted after a slow change in increasing indication before and after the indicator is lightly tapped. The procedure shall be repeated for decreasing indications. This test may be conducted at any point on the scale at the discretion of the Inspector. The maximum observed difference in reading shall not exceed 7.0°C. This test may be conducted in conjunction with the test specified herein for Scale error at room temperature.

4.5.5 Insulation resistance.- A potential of 200 volts, dc, shall be applied to each of the electrical receptacle pin contacts and the shell, shielding, or any other metal part of the indicator case. The insulation resistance measured between each of the electrical receptacle pin contacts and the shell, shielding, or any other metal part of the indicator case shall be not less than 20 megohms.

4.5.6 Sealing.- The indicator shall be immersed in a suitable liquid, such as water. The absolute pressure of the air above the liquid shall then be reduced to approximately 1 inch of mercury and maintained for 1 minute, or until air bubbles substantially cease to be given off by the liquid, whichever is longer. The absolute pressure shall then be increased to 2-1/2 inches of mercury. Any bubbles coming from within the indicator case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as a leak. A helium leak detector or other means of test equal or superior in sensitivity to the immersion test method described above may be used upon approval by the procuring activity.

4.5.7 Reduced voltage operation.- The indicator shall be tested with an applied voltage of 22.5 volts. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in table I.

4.5.8 Scale error at low temperature.- The indicator shall be subjected to a temperature of -55° ±2°C (-67° ±3.6°F) for a period of 4 hours. At the end of this period and while still at the low temperature, the indicator shall be subjected to the test specified herein for Scale error at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

4.5.9 Scale error at high temperature.- The indicator shall be subjected to a temperature of 70° ±2°C (158° ±3.6°F) for a period of 4 hours. At the end of this period and while still at the high temperature, the indicator shall be subjected to the test specified herein for Scale error at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

4.5.10 Open and short circuit.- With the test voltage applied to the indicator, the resistance bulb circuit shall be open-circuited five times and short-circuited five times, causing the pointer to go off scale in both directions. Following the test, the indicator shall be subjected to the test specified herein for Scale error at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

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4.5.11 Resistance bulb current.- A total resistance of 100 ohms shall be applied across the bulb circuit terminals of the indicator, the test voltage applied, and the bulb circuit current measured. The current consumed in the resistance bulb circuit shall not exceed 22 milliamperes.

4.5.12 Vibration error.- The indicator shall be mounted on a vibration stand so that a line joining the centers of the two lower mounting holes is in a horizontal plane and the plane of the mounting lugs in a vertical position. The indicator shall be subjected to vibration with an amplitude between 0.009 and 0.011 of an inch at frequencies from 500 to 3,000 cpm. Test voltage and resistance shall be applied to the indicator to provide an indication. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed 7.0°C. The difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 7.0°C. Pointer oscillation shall not be considered as pointer variation. This test may be conducted at any indication at the discretion of the Inspector.

4.5.13 Vibration failure.- The indicator shall be mounted on the vibration stand as specified herein for Vibration error test. The indicator shall be subjected to vibration with an amplitude between 0.018 and 0.020 of an inch for a 3-hour period. The vibration frequency shall be varied uniformly from 500 to 3,000 cpm and return once each hour during the test. This test may be conducted at any indication at the discretion of the Inspector. Following the 3-hour period, the indicator shall be subjected to the test specified herein for Scale error at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

4.5.14 Magnetic effect.- The indicator first not operating and second operating on full scale deflection shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The deflection of the compass at any of the specified positions shall not exceed 5 degrees.

4.5.15 Response time.- Voltage and resistance sufficient to produce momentary two-thirds full scale deflection shall be applied to the indicator. The length of time in seconds required for the pointer to come to rest shall be noted. This test shall be repeated several times and the average value taken as the response time of the indicator. The pointer shall be considered as having come to apparent rest when it has reached within ±1 percent of the scale length of the actual rest point. The response time shall not exceed 3 seconds.

4.5.16 Damping factor.- The test voltage shall be applied to the indicator and two suitable adjustable resistances connected in series across the resistance bulb connections. A short circuiting switch shall be connected across one of the variable resistances. The other variable resistance shall be so adjusted that the pointer reads at the lowest scale graduation when the first variable resistance is short-circuited. The first variable resistance shall then be so adjusted that the pointer momentarily deflects to the top scale graduation when the short circuit is relieved. The maximum momentary deflection shall be taken and recorded. After the pointer has come to rest, the steady deflection shall be taken and recorded. The damping factor shall be taken as the ratio of the steady deflection in angular degrees to the difference between the maximum momentary deflection and the steady deflection in angular degrees. The damping factor shall be not less than 1.6.

4.5.17 Power consumption.- The power consumption of the indicator shall be measured with the test voltage applied and sufficient test resistance to provide full scale deflection. The power consumption of the single indicator shall not exceed 1.5 watts, and the dual indicator shall not exceed 3.0 watts.

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4.5.18 Extreme low temperature exposure.- The indicator shall be subjected to a temperature of -65° ±2°C (-85° ±3.6°F) for a period of 48 hours. While still at this temperature, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The temperature shall then be raised to -55° ±2°C (-67° ±3.6°F) and the indicator subjected to this temperature for a period of 24 hours. The indicator shall then be subjected to the test specified herein for Scale error at low temperature, and the scale errors at the test points shall not exceed the tolerances specified in table I. The indicator shall then be returned to room temperature for a minimum period of 4 hours and then subjected to the test specified herein for scale errors at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

4.5.19 High temperature exposure.- The indicator shall be subjected to a temperature of 70° ±2°C (158° ±3.6°F) for a period of 24 hours. While still at this temperature, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The indicator shall then be returned to room temperature for a minimum period of 4 hours and then subjected to the test specified herein for Scale error at room temperature. The scale errors at the test points shall not exceed the tolerances specified in table I.

4.5.20 Altitude.-

4.5.20.1 High altitude.- The indicator shall be subjected to a maximum pressure of 0.82 inches of mercury absolute, at room temperature for a continuous period of 1 hour. While at this pressure, the indicator shall be operated throughout its range to determine any malfunctioning. There shall be no damage to the indicator as a result of this test.

4.5.20.2 High altitude - Low temperature.- The indicator shall be subjected to a pressure of 3.44 ±.08 inches of mercury absolute, and a temperature of -55° ±2°C (-67° ±3.6°F) for a continuous period of 4 hours. While at this pressure and temperature, the indicator shall be operated throughout its range to determine any malfunctioning. There shall be no damage to the indicator as a result of this test.

4.5.21 Thermal shock.- The indicator shall be subjected to eight cycles of exposure to water at 85° ±2°C (185° ±3.6°F) and 5° ±2°C (41° ±3.6°F) without evidence of moisture penetration or damage to coating or enclosure. Each cycle of the test shall consist of immersing the indicator in water at 85°C (185°F) for a period of 30 minutes and then within 5 seconds of removal from the bath, the indicator shall be immersed for a period of 30 minutes in the other water bath maintained at 5°C (41°F). This cycle shall be repeated continuously, one cycle following the other, until eight cycles have been completed. Following this test, the indicator shall be subjected to the test specified herein for sealing. No indicator leakage shall occur as a result of this test.

4.5.22 Salt spray.- The indicator shall be subjected to a salt spray in accordance with Specification QQ-M-151 for a period of 50 hours. Following this test, this indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The indicator shall then be subjected to the test specified herein for insulation leakage, and the measured resistance shall be a minimum of 20 megohms. The indicator shall then be subjected to the test specified herein for Scale error at room temperature, and the scale error at the test points shall not exceed the tolerances specified in table I. The indicator shall be examined, and there shall be no evidence of corrosion or deterioration which will affect subsequent operation. The external surface of the indicator may be washed with distilled water and air-dried prior to conducting the insulation leakage test.

4.5.23 Mounting lugs.- The indicator case with the mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no support. A suitable pin shall be inserted through the hole in the mounting lugs and attached to the pullstrap in the stationary head of the machine. A load of 175 pounds

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shall be applied for 1 minute to each lug along the lug axis and toward the front of the indicator. The lugs shall withstand the applied load without fracture.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation and packaging.- The preservation and interior packaging shall be in accordance with Specification MIL-P-6064, group X, in a reusable metal container in accordance with Specification MIL-C-6055.

5.3 Packing.- The exterior packing shall be in accordance with Specification MIL-P-6064.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Indicators; Temperature, Electrical-Resistance, -50° to +300°C, AN Part No., Manufacturer's Part No.

6. NOTES

6.1 Intended use.- The indicators covered by this specification are intended for use on aircraft to indicate temperatures in the range of -50° to +300°C, such as cylinder head.

6.2 Hermetic seal.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.3 Ordering data.- Requisitions, contracts, and orders should specify the AN part number of the indicator desired.

6.4 Provisions for Qualification tests.- In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Aeronautics, Navy Department, Washington 25, D.C., the activity responsible for qualification, and arrange to have the products that they propose to offer to the Army or the Navy, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the above designated activity responsible for qualification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy - Bureau of Aeronautics

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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